

What is the price of a model’s eye?: Understanding How VLMs Answer Multiple Choice Questions in Multimodal Settings

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Abstract

MCQA is a key competence of vision-language models that is tested by a variety of mainstream benchmarks. However, these models can have a wide range of changes in performance when the task is diversified even slightly. The question we are asking is: *how do mainstream vision-language models perform on formatted multiple-choice question answering tasks?* Using a counterfactual dataset derived from CLEVR scenes, we set out to evaluate and understand the models’ internal representations by employing methods like layer-wise activation patching, individual attention head localization, and vocabulary projection for answer choices to localize key hidden states and activation patterns that encode information about predictive tokens. Our experiments so far suggest that visual information is available early in image tokens, passes through option-content semantic representations in the middle layers, and reach the answer position and symbols in the late layers, mainly through attention. We find that all answer labels show a late increase in the logits, while a sparse set of attention heads separates the correct label. We also observe that common answer alphabets are more robustly promoted in vocabulary space than rare symbols. Our ongoing work will extend this analysis to other model families and to vision specific experiments separating visual-grounding features from symbol-binding features if time permits.

1 Introduction

Multi-choice question answering (MCQA) has been a popular task for evaluating LMs (Wiegreffe et al., 2025), and has been gaining traction in the space of vision-language models recently (Meng et al., 2024). A prominent example is the MMMU benchmark (Yue et al., 2024) and its successor (Yue et al., 2025) which evaluates multimodal models on massive multi-discipline tasks on a large number of open-source LMMs as well as proprietary models.

While LMs have been evaluated comprehensively in this domain, VLMs are yet to be probed in depth. In fact, its successor (Yue et al., 2025), removes the questions that text-only models can answer, expands the number of answer choices and adds a vision-only setting, under which the performance of VLMs dropped significantly. At the same time, prior work shows that text-only models are not robust to seemingly inconsequential format changes, such as shuffling the position of the correct answer or changing the symbols the answers are mapped to (Wiegreffe et al., 2025).

Our focus is not on whether the model answers correctly but on which layer and which attention heads produce that answer. While benchmarks like MMLU and MMMU are widely trusted, prior work shows that text-only models can be sensitive to superficial changes, such as answer order or label symbols (Pezeshkpour and Hruschka, 2024; ?; Khatun and Brown, 2024; Zheng et al., 2024).

Building upon the work of (Wiegreffe et al., 2025), we aim to understand how VLMs form their predictions, at which layers they decide on it, particularly on the format of MCQA, which is important for reliability reasons.

Our current efforts begin with smoke-testing of the models from families such as Qwen (Bai et al., 2025a), PaliGemma (Beyer et al., 2024), LLaVA-OneVision (Li et al., 2024), and InternVL (Wang et al., 2025), to select the best models from each, and a single dataset, CLEVR-MCQ-4 built on CLEVR (Johnson et al., 2016) with synthetic mcqas that we create. Our preliminary findings on Qwen2.5-VL-7B (Bai et al., 2025b) and InternVL3.5-8B (Wang et al., 2025) are:

1. When models are correct, all answers receive a shared logit increase in later layers promoting all labels together before separating the correct label from the rest.
2. Activation patching suggests a staged route:

the visual information available causally in image tokens early on, carried through option-content representations in the middle layers, and written to the final token in late layers.

3. The late transfer of information is mainly driven by attention rather than MLPs and a sparse set of attention heads accounts for a large share of the attention effect.
4. Robustness of common alphabets over rare symbols is quite evident as even when the correct rare symbol wins under forced choice, much of the full-vocabulary probability goes to other tokens.

2 Related Work

Option order and output symbols can alter MCQA predictions despite unchanged answer alternatives (Zheng et al., 2024). Atabuzzaman et al. (2025) characterize token and position preferences in fine-grained visual recognition, while Zeno et al. (2025) examine selection bias in spatial MCQA. These behavioral findings motivate locating the decoder components connecting visual answer content to the emitted label.

In Answer, Assemble, Ace (AAA), Wiegrefe et al. (2025) combine vocabulary projections and activation patching under option-position and alphabet changes to distinguish answer representations from output-symbol promotion in text-only MCQA. Wong et al. (2026) use trained probes and probe-aligned interventions, finding answer-relevant information at option boundaries and stronger symbol-binding effects near the output. Correct-letter head analyses localize answer-label contributions, while path patching of symbol-biased heads localizes sensitivity to output alphabets; these are complementary roles within answer-to-label mapping (Lieberum et al., 2023; Yang et al., 2025). Together, these findings motivate partly separable content and label processes, although readout accessibility alone does not establish where answer information is first computed or causally used (Wiegrefe et al., 2025; Wong et al., 2026). Patching outcomes depend on corruption and metrics, while estimates vary across inputs and component selection can amplify structural instability (Zhang and Nanda, 2024; Méloux et al., 2026).

In VLMs, activation swaps with downstream context cached can redirect object-word associations (Saravanan et al., 2025), while controlled

swaps and head interventions provide evidence for spatial indexing and feature retrieval (Assouel et al., 2026). These object and spatial binding targets differ from mapping an MCQA answer to its output label. Golovanevsky et al. (2025) introduce NOTICE, separating semantic image-pair corruption from candidate-word replacement and patching multimodal modules and heads to test image- and text-dependent effects. Palacios et al. (2026) report intermediate semantic separation and probe accessibility before late vocabulary-letter alignment in VLMs, relating these patterns to cross-modal representation alignment during visual instruction tuning. Their representation geometry and probes describe separation and accessibility, while image-attention knockout and layer skipping test functional importance across instruction-tuning stages (Palacios et al., 2026).

We adapt AAA’s answer-to-label analysis and NOTICE’s semantic image interventions to controlled visual MCQA (Wiegrefe et al., 2025; Golovanevsky et al., 2025). Relative to probe-based and layer-level accounts, our frozen-checkpoint study combines untuned vocabulary readouts with direct head interventions (Wong et al., 2026; Palacios et al., 2026). With visual content and option position independently controlled, we investigate which decoder states and attention heads causally mediate image-dependent answer changes and position-dependent label changes. Sparse-head confirmation concerns A/B/C/D position contrasts and tests whether a frozen head set retains position-transfer effects on disjoint semantic pairs relative to layer-matched random controls.

3 Exploratory Study

3.1 Controlled task and the two-signal observation

We began with Qwen2.5-VL-3B and 64 pairs of simple four-shape images. Within a pair, layout and colour counts are fixed while the queried object’s colour changes. The answer content, correct option position, and label alphabet can therefore be varied independently. Across 2,560 prompts, we measured *restricted accuracy* among the four displayed labels, whether the full-vocabulary top token was a valid label, and the total probability mass on the four labels (Table 1). These are different quantities: a model may win a forced four-way comparison even when it would naturally emit another token.

For A/B/C/D, every candidate logit rises sharply

Labels	Restr. (%)	Valid top (%)	Mass (%)	> 90% at
A/B/C/D	100.00	100.00	100.00	28
1/2/3/4	100.00	100.00	98.27	32
Q/Z/R/X	97.50	69.69	62.84	33
!/@/#/\$	95.94	25.47	30.83	35

Table 1: Qwen2.5-VL-3B pilot. “Valid top” uses the full vocabulary; mass is the full-vocabulary probability on the four labels. The last column is the first residual stage whose mean restricted correct-label probability exceeds 90%.

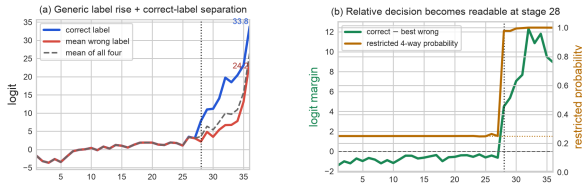


Figure 1: All four familiar labels share a late rise in the upper layers, while the correct label separates on top of it. Stage 0 being omitted for readability. The logit lens is observational.

near the output. This shared rise is consistent with a generic signal that the next token should be an answer label. The correct label receives an additional relative increase and separates from the other candidates (Figure 1). This decomposition is descriptive, not a causal explanation of the shared rise.

Correct-symbol separation is prior and strongest for A/B/C/D, later for numbers, and latest and weakest for rare letters and punctuation. Q and ! account for all 42 unfamiliar-alphabet forced-choice errors, whereas A and 1 are perfect; the effect is therefore not a general first-position failure. Default A/B/C/D directions can also dominate before the requested alphabet takes over late.

3.2 Causal pilot and working hypothesis

Activation patching then tested whether a donor state could steer a recipient. Table 2 reports representative held-out effects. Normalized recovery is 0 for no movement and 1 for full donor transfer.

The pilot shows a testable route: visual content early, candidate states in the middle, and a late position-to-symbol write. Image and option patches too show why late vocabulary readability must not be mistaken for the presence of answer information.

Site	Representative result and supported reading
Image, 0-16	Recovery 0.92, 0.93, and 0.87: visual answer information is causally present early.
Options, 20-25	Content recovery peaks near 0.48: answer information later appears in candidate text.
Labels, 26	Position recovery is about 0.46: option identity associated with a displayed label.
Final token, 27	Symbol recovery is 0.76; attention 0.68 versus MLP 0.27, indicating an attention-mediated late update.
Final token, 33	Content recovery is 0.95; attention 0.55 versus MLP 0.03, while the answer becomes vocab-aligned.

Table 2: Representative causal pilot results. The sites support a sequence, not a proven edge-by-edge circuit.

4 Dataset and Model Selection

4.1 CLEVR-MCQ-4

The proposal considered MMMU, a heterogeneous benchmark of expert-level multimodal questions (Yue et al., 2024). For repeatedly patched sub-10B models, moderate baseline accuracy makes a failed intervention ambiguous. We therefore built CLEVR-MCQ-4 from CLEVR’s diagnostic scenes and functional reasoning structure (Johnson et al., 2016).

The release covers nine reasoning families with type-matched distractors. It contains three answer alphabets and five causal contrast types: position-only, content-only, content-and-position, identity, and unrelated random pairs. Splits are scene-disjoint and fixed before analysis (Table 3). Figure 2 illustrates the intended decomposition.

4.2 Model screening

The same 24-item smoke profile tested three checkpoints from each of four families: Qwen, InternVL, LLaVA-OneVision, and PaliGemma (Bai et al., 2025b; Wang et al., 2025; Li et al., 2024; Beyer et al., 2024). The screen checked adapter and intervention compatibility, task performance, image dependence, and available causal pairs. It guides selection rather than serving as a leaderboard be-

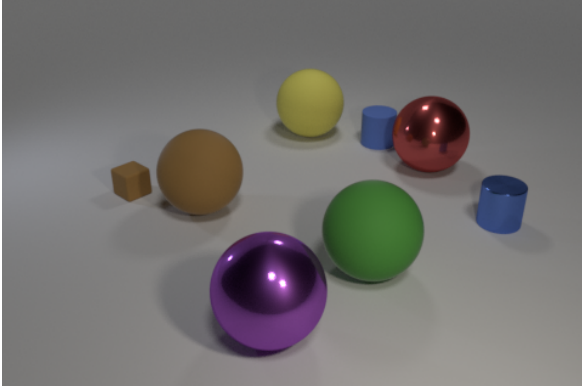


Figure 2: Example task: identify the brown object through a visual relation, locate “brown” among the options, and emit its assigned label A.

Split	Items	Use
Calibration	120	Dataset checks only
Screening	300	Optional model screen
Behaviour	256	Accuracy and lenses
Discovery	320	Select layers and heads
Confirmation	256	Frozen held-out tests
Total	1,252	17,328 prompt variants

Table 3: Frozen CLEVR-MCQ-4 release. Discovery has 160 pairs and confirmation has 128 pairs.

cause each estimate uses only 24 items.

Qwen2.5-VL-7B and InternVL3.5-8B offered the strongest combination of image-dependent performance, calibrated interventions, sufficient pairs, and feasible size. The 24-item estimates remain screening evidence and are not pooled with the causal runs.

The sequential full InternVL discovery-to-confirmation sweep was projected from measured throughput to require roughly 50-60 hours. To obtain a timely cross-model test without reducing layer resolution or interrupting that run, we used a separate report track with 32 pair-balanced counterfactual cases per contrast in each disjoint split, rather than 64. It retained the same frozen release, checkpoint, prompts, gates, hooks, all 36 residual layers, and discovery-before-confirmation protocol; its estimates are therefore a smaller matched replication, reported as preliminary until the full sweep is audited.

5 Core Methodology

Core-7 combines four interventions with three descriptive readouts. For donor and recipient exam-

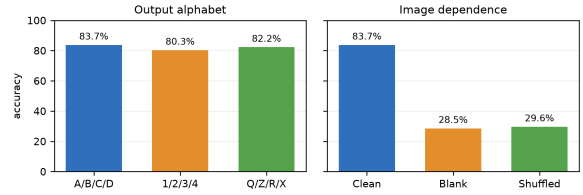


Figure 3: Accuracy is similar across output alphabets but collapses when useful visual evidence is removed or mismatched.

ples, normalized recovery is $NR = (s_{\text{patched}} - s_{\text{recipient}}) / (s_{\text{donor}} - s_{\text{recipient}})$ for a predeclared logit difference s . We also report flip rate, the fraction of patched predictions that become the donor answer. Recovery can change without a categorical flip.

All selection decisions use discovery data and are frozen before confirmation. The full Qwen workflow uses 64 pairs and 28 decoder layers. The preliminary InternVL track uses 32 pair-balanced cases per contrast and all 36 layers; token groups are sampled every second layer, and frozen head sets are compared with ten exact-size, layer-matched random sets. Reported sample sizes remain attached to their experiments; smoke, preliminary, and full results are not pooled.

6 Results

6.1 Behavioural foundation

The 256-item Qwen behavioural set establishes that the model solves the task using visual evidence. Accuracy is 83.7% with A/B/C/D, 80.3% with numbers, and 82.2% with rare letters. It falls to 28.5% with a blank image and 29.6% with a shuffled image (Figure 3). Logical OR remains difficult at 44.7%, so subsequent causal averages use pairs for which donor and recipient are both answered correctly. On the matched 24-item screen, InternVL reaches 100.0% A/B/C/D accuracy and loses 78.1 percentage points when images are shuffled. This smaller estimate establishes feasibility and image dependence; it is not pooled with Qwen’s 256-item result.

6.2 M1 and M6: when and where the answer transfers

Method. M1 patches the final answer-token state from a donor into a recipient at each layer. M6 instead patches token groups to identify the carrier of the effect.

Family	Model (B)	Acc.	Gap	D/C	Family	Model (B)	Acc.	Gap	D/C
Qwen	Qwen2-VL (2.2)	.7292	.4167	6/4	LLaVA-OV	0.5B-SI (0.9)	.4063	.1771	1/2
Qwen	Qwen2.5-VL (3.8)	.8438	.5729	6/6	LLaVA-OV	7B-SI (8.0)	.7292	.4479	3/4
Qwen	Qwen2.5-VL (8.3)	.9375	.6458	6/6	LLaVA-OV	7B-OV (8.0)	.6458	.3542	6/6
InternVL	3.5-1B (1.1)	.8750	.5417	5/4	PaliGemma	3B Mix (2.9)	.4271	.2292	4/3
InternVL	3.5-2B (2.3)	.9063	.6667	4/5	PaliGemma	2-3B Mix (3.0)	.5000	.2604	2/3
InternVL	3.5-8B (8.5)	1.0000	.7813	6/5	PaliGemma	2-10B Mix (9.7)	.5833	.3958	2/3

Table 4: Condensed 24-item model screen. Accuracy is restricted A/B/C/D accuracy; Gap is clean minus shuffled-image accuracy; D/C gives available discovery/confirmation pairs. Worst-position accuracy was also checked. Bold marks the provisional family representative.

	Plain-language question	Measurement and evidential role
M1	When can a donor answer steer the output?	Patch the final-token state by layer; random pairs test whether transfer is contrast-specific.
M2	Does attention or the MLP matter more?	Patch and ablate their updates separately; vocabulary projections remain descriptive.
M3	Do a few attention heads dominate?	Select heads on discovery, freeze them, and compare with layer-matched random sets on confirmation.
M4	When is the answer label readable?	Project every residual stage to the vocabulary; this measures accessibility, not causal origin.
M5	Is content readable before the symbol?	Compare answer-word and answer-label projections through depth.
M6	Which token groups carry the effect?	Patch image, question, option, label, and final-token groups under controlled contrasts.
M7	Do unfamiliar labels change the route?	Repeat behaviour and readouts across alphabets, including default A/B/C/D directions.

Table 5: Core-7. M1, M2, M3, and M6 intervene on activations. M4, M5, and M7 are observational readouts and are interpreted accordingly.

Results. M1’s final-token transfer crosses its frozen threshold at layer 24 on both discovery and confirmation. For position-only pairs, the donor-label flip rate rises from 2.7% at layer 23 to 73.5% at 24, 84.9% at 25, and 100.0% by 27. However, unrelated random pairs transfer almost as strongly (Figure 4). M1 therefore establishes a late transferable donor state, but does not by itself isolate a contrast-specific binding code.

M6 supplies the carrier information. Under content contrasts, image-token recovery is 101.0% through layers 0-8, falls to 24.0% at layer 16 and 6.0% at layer 24, while option-content recovery reaches 38.0% at layer 16 and peaks at 43.0% at

layer 18. In content-and-position confirmation pairs, final-token patches primarily transfer the donor position/symbol: position flips rise from 74.0% at layer 24 to 100.0% at 27. Together, M1 and M6 support an early image-token carrier, a middle option-content carrier, and a late output-symbol state.

InternVL reproduces this ordering on its disjoint confirmation split: position transfer crosses threshold at layer 29 and content transfer at 31. At layer 29, position recovery is 54.2% and the donor-position flip rate is 71.9%. Its random-pair recovery is again stronger (68.3%), so the same interpretive restriction applies. M6 also begins with strong

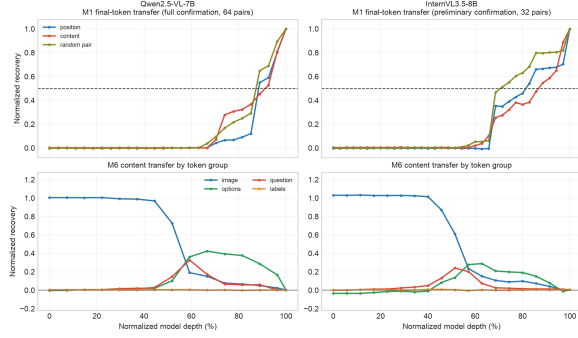


Figure 4: M1 and M6 on held-out pairs. Top: final-token recovery for position, content, and random donors. Bottom: content recovery by patched token group. Depth is normalized across Qwen’s 28 layers (64 pairs) and InternVL’s 36 layers (32 preliminary pairs).

image-token recovery and peaks later in question and option-content tokens. The shared result is an image-to-text-to-symbol route; Qwen’s late transition is sharper, whereas InternVL’s develops more gradually.

6.3 M2: attention and MLP contributions

Method. M2 patches and resampling-ablates the attention and MLP updates separately. A direct vocabulary projection is included only to describe what each update aligns with.

Results. Attention-output patch recovery peaks at layer 24: 44.1% on discovery and 43.9% on confirmation, versus an MLP peak near 10.0%. Resampling the layer-24 attention update reduces the correct-label margin by 5.0 logits, compared with 0.6 for the MLP. At the final layer, MLP ablation has the larger effect (1.5 versus 0.7 logits). The projection in Figure 5 shows a sharp layer-24 attention write to the correct-minus-wrong margin, whereas late MLP updates align more with a shared rise in candidate-label logits. The causal evidence comes from patching and ablation, not from the projection alone.

InternVL shows the same component division. Confirmation attention recovery peaks at 35.0% at layer 24, compared with a 4.6% MLP peak; resampling the layer-24 attention update reduces the answer margin by 5.1 logits, while the MLP change is -0.1 . Thus the main discriminative write is attention-mediated in both models, while the projection-based generic-label interpretation remains descriptive.

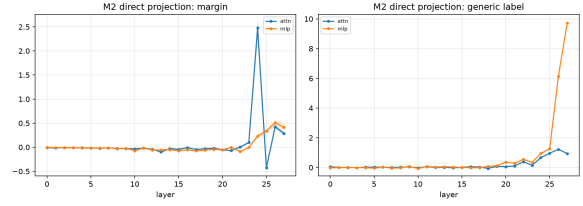


Figure 5: M2 direct projection. Attention has its largest discriminative write near layer 24 (left); late MLP updates show the largest generic candidate-label write (right).

6.4 M3: important heads, with model-dependent sparsity

Method. M3 ranks heads with equal pair weight, freezes each top- k set, and evaluates it on a disjoint confirmation split. Qwen uses 64 pairs per split and InternVL uses 32; exact-size, layer-matched random sets form each null distribution.

Results. Confirmation recovery rises from 11.6% for one head to 24.4% for two, 41.7% for four, 68.4% for eight, and 89.5% for sixteen. Four heads recover 52.7% of the all-head reference and exceed the matched-random 95th percentile of 6.9%, satisfying the predeclared sparse-set criterion (Figure 6). This establishes a reproducible, concentrated causal effect, not a complete or uniquely necessary circuit.

InternVL’s frozen sets also have a clear held-out effect: one, four, and sixteen heads recover 20.4%, 43.1%, and 92.4%, and every tested top- k set exceeds its matched-random 95th percentile. The sparsity classification is not stable, however. Four heads recover 49.9% of the all-head effect in discovery but 50.8% in confirmation. The predeclared verdict therefore moves from “distributed” to “sparse” at its 50% boundary. We infer selected head concentration in both models, but stable sparsity only in Qwen.

6.5 M4, M5, and M7: readable content and output symbols

Method. M4 follows the answer labels through depth, M5 compares the semantic answer labels through depth, M5 compares the semantic answer word with the printed label, and M7 repeats the readout for different output alphabets. All three are vocabulary projections: they measure linear readability in the final vocabulary basis, not causal use.

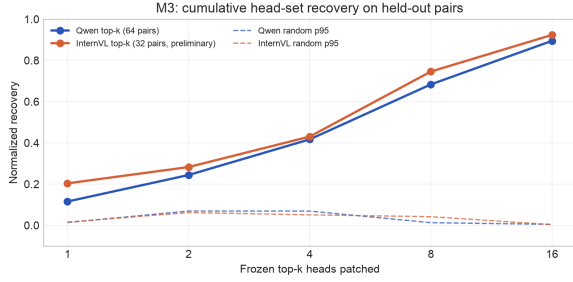


Figure 6: M3 confirmation recovery for frozen top- k head sets. Dashed curves show layer-matched random 95th percentiles. InternVL’s selected-head effect is clear, but its sparse/distributed verdict does not replicate.

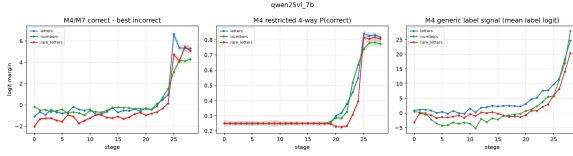


Figure 7: M4/M7 readouts. Correct-label separation, restricted confidence, and the shared candidate-label lift all emerge late.

Results. The familiar-letter margin becomes stably positive around stages 22-23 of 28; numbers follow at 23 and rare letters at 24. All candidate labels also receive a shared late lift (Figure 7). For rare-label prompts, the requested alphabet holds only 12.6% of the combined requested/default eight-way mass at stage 24, then 82.8% at 25. The final-token answer word does not become stably positive until stage 27, after the answer symbol is already readable (Figure 8). M6 prevents this late content-word readout from being mistaken for the first presence of answer content: content has earlier causal carriers in image and option tokens.

The available matched InternVL lens evidence is the six-pair smoke analysis, not the 32-pair causal track. It nevertheless gives the same qualitative ordering: A-D labels become stably readable at stage 25 of 36, the requested rare alphabet overtakes default A-D directions at stage 30, and the content word becomes readable at stage 32. Across models, output-symbol alignment therefore precedes final-token content-word alignment, while M6 shows that answer content is causally active at earlier non-final token sites.

7 Cross-model Interpretation

Both models support the same broad computation: visual states carry answer content early, textual option states become useful in the middle, and

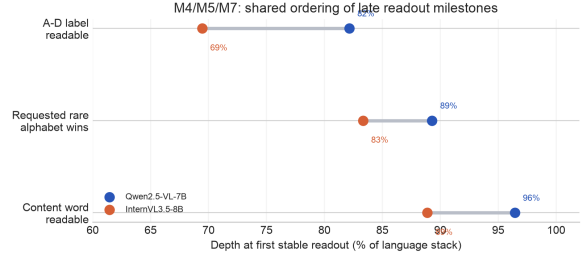


Figure 8: Matched M4/M5/M7 readout milestones, normalized by language-stack depth. InternVL milestones come from the six-pair smoke lens and remain preliminary. The shared result is the ordering, not identical absolute depth.

late attention has the largest causal effect on the emitted position/symbol. They also share strong random-pair portability and a late switch from default A/B/C/D directions into the requested output alphabet. The main differences are quantitative: InternVL’s late transfer is more gradual, and its sparse-head verdict is threshold-sensitive at 32 pairs. Agreement therefore strengthens the staged VLM-level account, while the unstable head result identifies the claim most in need of the continuing full InternVL sweep.

8 Conclusion

Qwen and preliminary InternVL evidence support a staged account: answer-relevant information is causally present in image tokens early, textual option states become useful in the middle, and late attention strongly affects the selected output symbol. Frozen head sets have substantial held-out effects in both models, although only Qwen currently supports a stable sparse verdict. Vocabulary projections reveal late label readability and alphabet switching rather than the first computation of answer content. The account is deliberately narrower than a complete circuit claim: strong random-pair transfer shows that late donor-state portability is not specific to the controlled semantic contrast.

Limitations

The study covers two 8B-class VLMs on one controlled dataset, and both language backbones are Qwen-lineage. InternVL causal estimates use 32 pair-balanced cases per contrast and remain preliminary until the longer full sweep is complete and audited. Some question families remain difficult, and causal analyses condition on correct donor/recipient pairs. Patching supports effects

only for the tested source, target, site, and metric; vocabulary projections remain correlational. Neither model’s selected heads constitute a complete or unique circuit.

9 Timeline

Dates	Milestone
<i>Mid-submission: October 31st</i>	
Oct 1 – Oct 12	Complete the remaining core experiments across all four model families.
Oct 13 – Oct 24	Conduct stretch experiments, prioritizing cross-image patching and failure-mode analysis, followed by remaining experiments if time permits.
Oct 25 – Oct 31	Consolidate results, prepare figures, and write the final report.
<i>Final submission: October 31</i>	

Table 6: Project Timeline

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